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CODE:

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# Implementation of Linear Regression Algorithm

import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Sample Dataset
# X = Number of hours studied
# y = Marks obtained

X = np.array([
    [1], [2], [3], [4], [5],
    [6], [7], [8], [9], [10]
])

y = np.array([
    35, 40, 50, 55, 60,
    65, 70, 75, 85, 90
])

# Split the dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Create Linear Regression Model
model = LinearRegression()

# Train the Model
model.fit(X_train, y_train)

# Predict Test Data
y_pred = model.predict(X_test)

# Calculate Mean Squared Error
mse = mean_squared_error(y_test, y_pred)

# Calculate R-squared Score
r2 = r2_score(y_test, y_pred)

# Display Results
print("LINEAR REGRESSION RESULTS")
print("-----")

print("\nCoefficient (Slope):", model.coef_[0])
print("Intercept:", model.intercept_)

print("\nActual Values:", y_test)
print("Predicted Values:", y_pred)

print("\nMean Squared Error:", mse)
print("R-squared Score:", r2)
```

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# Predict marks for a new input
hours = np.array([[11]])

predicted_marks = model.predict(hours)

print("\nHours Studied:", hours[0][0])
print("Predicted Marks:", predicted_marks[0])
```

OUTPUT :

LINEAR REGRESSION RESULTS

Coefficient (Slope): 5.775862068965519
Intercept: 30.732758620689648

Actual Values: [85 40]
Predicted Values: [82.71551724 42.28448276]

Mean Squared Error: 5.218861474435167
R-squared Score: 0.9896911378282762

Hours Studied: 11
Predicted Marks: 94.26724137931035